

CLINICAL TRIALS

Endovascular Thrombectomy in Patients With Largest Baseline Infarcts (ASPECTS 0–2): An Ancillary Analysis of the LASTE Trial

Bertrand Lapergue¹, MD, PhD; Caroline Arquizan², MD; J.F. Albucher³, MD; Christophe Cognard⁴, MD, PhD; Hilde Henon⁵, MD; Nasreddine Nouri⁶, MD; Benjamin Gory⁷, MD, PhD; Sebastien Richard⁸, MD, PhD; Gauthier Marnat⁹, MD; Igor Sibon¹⁰, MD, PhD; Federico Di Maria, MD; Mariam Annan, MD; Grégoire Boulouis¹¹, MD; Pere Cardona¹², MD; Julien Labreuche¹³, BST; Vincent Costalat, MD, PhD; Tudor G. Jovin¹⁴, MD; for the LASTE Trial Investigators*

BACKGROUND: Endovascular thrombectomy (EVT) has demonstrated benefits in patients with ischemic stroke and large-vessel occlusions, but its efficacy and safety in patients with the largest baseline infarcts (Alberta Stroke Program Early Computed Tomography Score [ASPECTS] 0–2), remain controversial. This study aimed to evaluate the efficacy and safety of EVT in early presenting patients with ASPECTS 0 to 2 compared with medical care alone.

METHODS: This post hoc analysis of the multicenter, randomized LASTE trial ([Large Stroke Therapy Evaluation]; patients ≤80 years, ASPECTS 0–5, proximal anterior circulation large-vessel occlusion, randomization within 6.5 hours of last known well) evaluated the subgroup with ASPECTS 0 to 2, representing those with large baseline infarcts without an upper size limit, randomized to receive EVT plus medical care or medical care alone. Primary outcomes included the distribution of the modified Rankin Scale score at 90 days. Secondary outcomes included mortality, infarct volume growth at 24 hours, the incidence of symptomatic intracranial hemorrhage and modified Rankin Scale score at 180 days. The LASTE trial was conducted and reported in accordance with the CONSORT guidelines (Consolidated Standards of Reporting Trials).

RESULTS: Median age was 72 years and 55.8% were women. Among the 181 patients with ASPECTS 0 to 2 (median core volume 156 mL [25th–75th percentiles, 121–204 mL]), at 90 days after randomization, EVT improved functional outcomes (generalized odds ratio, 1.81 [95% CI, 1.32–2.47]) and reduced mortality (38.4% versus 59.6%; relative risk, 0.64 [95% CI, 0.47–0.89]), which translated predominantly into an increase in the proportion of modified Rankin Scale score of 0 to 3 in the EVT group (31.4% versus 8.5%; relative risk, 3.69 [95% CI, 1.77–7.68]). A significant reduction in infarct growth volume was observed in the EVT group compared with medical care (mean difference, −70.3 mL [95% CI, −94.2 to −46.3]). Rates of symptomatic intracranial hemorrhage were 12.9% versus 4.5%, respectively (relative risk, 2.85 [95% CI, 0.94–8.60]).

CONCLUSIONS: EVT improves functional outcomes and reduces mortality in patients with ASPECTS 0 to 2. These findings support the concept that in patients aged <80 years presenting within early time window (6.5 hours) with unlimitedly large infarct (predominantly selected using magnetic resonance imaging), the infarct size in isolation should not be used to disqualify patients from endovascular treatment.

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GRAPHIC ABSTRACT: A [graphic abstract](#) is available for this article.

Key Words: acute ischemic stroke ■ infarction ■ intracranial hemorrhages ■ stents ■ thrombectomy

Correspondence to: Bertrand Lapergue, MD, PhD, Stroke Center Neurology Division, Hôpital Foch, 40 Rue Worth, Suresnes 92150, France. Email bertrand.lapergue@gmail.com

*A list of all LASTE Trial Investigators is given in the Appendix.

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Nonstandard Abbreviations and Acronyms	
ASPECTS	Alberta Stroke Program Early Computed Tomography Score
BMT	Best Medical Treatment
EVT	Endovascular Thrombectomy
mRS	modified Rankin Scale
sICH	symptomatic intracranial hemorrhage

The indications for endovascular thrombectomy (EVT) in acute ischemic stroke with large-vessel occlusion have expanded after the recent publication of 6 randomized controlled trials¹ supporting its efficacy and safety in patients with large-core infarct.^{2–7} However, trials largely included patients with a baseline Alberta Stroke Program Early Computed Tomography Score (ASPECTS) of 3 to 5, leaving the benefit of EVT in cases with the largest infarcts (ASPECTS 0–2) an ongoing subject of debate, especially concerning the futility and the hemorrhagic risk of EVT.

The LASTE trial (Large Stroke Therapy Evaluation) specifically addresses this gap by including patients with no restrictions on the upper limit of infarct size within 6.5 hours from last known well (ASPECT 0–5).³ By analyzing patients who were excluded from previous trials, this study contributes important data to inform future clinical decision-making. The time window has been restricted to 6.5 hours between symptoms and randomization, to achieve complete recanalization within the limit of 8 hours. This timing has been based on prior evidence indicating diminishing effect of reperfusion after ≈6.5 hours, and alignment with the time windows used in early window thrombectomy trials.⁸ The objective of this post hoc analysis of the LASTE trial was to assess the efficacy and safety of EVT in patients with ASPECTS 0 to 2 within 6.5 hours from last known well.

METHODS

The data that support the findings of this study are available from the corresponding author on reasonable request.

Study Design

The present study is a post hoc analysis of the LASTE trial. The LASTE trial was a multicenter, prospective, open-label, randomized controlled trial with blinded outcome evaluation, conducted at 24 hospitals in France and 6 hospitals in Spain from April 2019 through March 2023, which compared EVT plus best medical treatment (BMT) versus BMT alone in patients with large infarcts (ASPECTS 0–5) in the early therapeutic window designed to assess the efficacy of EVT in patients with large ischemic infarcts. In this present study, we specifically focused on the subgroup of patients with ASPECTS 0 to 2 at baseline, assessed by core laboratory evaluation.

Population

The trial protocol was approved by the Comité de Protection des Personnes EST-III (No. Comité Protection des Personnes: 18.10.02; No. RCB: 2018-A02102-53) and by the institutional review board at each participating center. Written informed consent was obtained from all patients or their legally authorized representatives.

Patients were eligible for inclusion in LASTE, if they were aged >18 years; had an ASPECTS value of 0 to 5 on compute tomography (CT) or magnetic resonance imaging (MRI), except for patients older than 80 years of age, for whom a baseline ASPECTS value of 4 or 5 was required; had an occlusion of the intracranial segment of the internal carotid artery or the proximal (M1) segment of the middle cerebral artery (or both); had a prestroke modified Rankin Scale (mRS) score of 0 or 1; had a National Institutes of Health Stroke Scale score of at least 6 (scores range from 0 to 42, with higher scores indicating more severe stroke); and could undergo randomization within 6.5 hours after the time since the patient was last known to be well. Patients with an unknown time of symptom onset could be included if they presented within 24 hours of last known to be well and the lesion on diffusion-weighted MRI had no corresponding lesion on fluid-attenuation inversion recovery imaging, a signature indicating that <4.5 hours had passed since stroke onset.⁹ Patients were excluded if there was evidence of intracerebral hemorrhage. Details of the inclusion and exclusion criteria in the LASTE trial have been previously reported.^{3,10}

Patients included in the LASTE trial were eligible for this current analysis when the core lab adjudicated ASPECTS was of 0 to 2 on baseline imaging. The ASPECTS was assessed by core laboratory evaluation.

Outcomes

The primary outcome was the shift toward better functional outcomes on the 90-day mRS score, combining scores of 5 and 6 into one category. The pooling of mRS score categories 5 and 6 for the ordinal primary outcome was prespecified in the LASTE statistical analysis plan. Secondary outcomes included the mRS score at 180 days after randomization, mRS score of 0 to 2, mRS score of 0 to 3 (independent ambulation) and proportion of patients with severe disability (mRS score 4–5) at 90 days and 180 days, early neurological improvement (defined as a decrease in the National Institutes of Health Stroke Scale score of ≥8 points from baseline at either 7 days or the time of hospital discharge); decompressive craniectomy rate, change in infarct volume at 24 hours

Safety outcome included 90-day mortality, early neurological worsening at 24 hours, any intracranial hemorrhage, symptomatic intracranial hemorrhage (sICH) at 24 hours, based on the Heidelberg classification.¹¹

Statistical Analysis

Analyses were based on all randomized patients with ASPECTS 0 to 2 as graded by core lab, in the allocated arm, after exclusion of patients due to consent withdrawal or other legal reasons. All analyses were done without missing value imputation. Distribution of mRS score (after pooling together mRS score of 5 and 6) at 90 days (primary outcome) and at 180 days were compared between thrombectomy and control group using the

Wilcoxon–Mann–Whitney *U* test and after determining that the proportional odds assumption could not be verified, treatment effect sizes were estimated using the generalized odds ratio by splitting ties equally between groups. For the primary outcome only, leave-one-out site analyses were done as well as subgroup analyses on prespecified key characteristics according to age (≤ 70 versus > 70 years), sex, use of intravenous thrombolysis, admission National Institutes of Health Stroke Scale score (< 20 versus ≥ 20), last known well to randomization time (0–4.5 versus ≥ 4.5 –6.5 hours), admission mode (Drip and Ship versus Mothership), and type of qualifying imaging modalities (MRI versus CT). Comparison of secondary outcomes were done using χ^2 test for binary outcomes, using a constrained longitudinal data analysis model for infarct volume change from baseline to 24 hours, and using a Fine and Gray regression model for the incidence of decompressive craniectomy. Treatment effect sizes were assessed in term of relative risk, baseline-adjusted mean difference, and subdistribution hazard ratio, respectively. Overall survival during the 180-day was estimated using Kaplan–Meier method and compared using a Cox proportional hazard; hazard ratio of all-cause mortality was reported as effect size. All statistical tests were 2-sided and $P < 0.05$ was considered statistically significant. Because of the potential for type 1 error due to multiple comparisons, findings should be considered as exploratory. Data were analyzed with SAS software version 9.4 (SAS Institute, Cary) and by using Gen-odds R package.

RESULTS

Among the 324 patients analyzed in the LASTE trial, 181 patients (56%) had an ASPECTS 0 to 2 at inclusion as graded by the core lab and were considered for inclusion in this post hoc analysis (Figure S1). These patients were enrolled from 29 of 30 LASTE centers (Figure S2), and 86 patients were randomized to the EVT+BMT arm group and 95 patients in the BMT arm. Baseline characteristics are reported in Table 1 and were well balanced according to the allocated arm. The median age was 72 years, 55.8% of patients were women. ASPECTS was graded as 0 in 34 patients (18.8%), 1 in 58 patients (32.0) and 2 in 89 patients (48.4%). The median baseline infarct (core) volume was 156 mL (25th–75th percentiles, 121–204 mL); 32.6% of EVT patients received intravenous thrombolysis (Table 1).

Intervention Details

EVT was performed in 84 of 86 patients allocated to the mechanical thrombectomy group, at a median time of 309 min (interquartile range, 230–381) after symptom onset; modified Thrombolysis in Cerebral Infarction grade $\geq 2b$ reperfusion was achieved in 88.1% ($n=74$) of patients (Table S1).

Efficacy

The distribution of the mRS score at day 90 (primary outcome measure) and at day 180 are presented in

Figure 1. We found a significant favorable shift in the distribution of 90-day mRS score in the EVT+BMT compared with the BMT group, with a generalized odds ratio of 1.81 (95% CI, 1.32–2.47; Table 2), signifying a 81% increased odds of less disability with EVT compared with control (number needed to treat, 3.5 [95% CI, 2.3–7.2]). Similar results were found at day 180 (generalized odds ratio, 1.69 [95% CI, 1.23–2.31]; Table 2; Figure 1) or in leave-one-out (center) sensitivity analysis (Figures S3 and S4) or in key subgroups (Figure 2).

Analyzing mRS score as binary outcomes generally favored EVT, with a relative risk (RR) for independent ambulation (mRS score, 0–3) of 3.69 (95% CI, 1.77–7.68) and 3.06 (95% CI, 1.58–5.93) at day 90 and 180, respectively. The percentage of patients with mRS scores of 0 to 2 at 90 days and 180 days are reported in the Table 2. EVT was associated with greater early neurological improvement (RR, 5.42 [95% CI, 1.93–15.21]).

The proportions of patients with severe disability (mRS score 4 and 5) at three and 6 months did not differ in the thrombectomy group compared with the control group (30.2% versus 31.9%, $P=0.81$; Table 2).

As shown in Figure 2, we found no evidence of treatment effect size heterogeneity in the primary outcome according to the prespecified LASTE subgroups.

Infarction growth rates on imaging between baseline and 24 hours, were significantly lower in the EVT+BMT arm compared with the BMT group (baseline adjusted mean difference in infarct growth volume of -70.3 mL [95% CI, -94.2 to -46.3]; Figure S5).

Safety

Mortality rates at 90 days were lower in the EVT+BMT group compared with the BMT group (38.4% versus 59.6%, RR, 0.64 [95% CI, 0.47–0.89]; Table 2), a difference that was similar at 180 days (Figure S4). The sICH rates at 24 hours (see Table S2 detailing all intracranial hemorrhage) in the EVT group compared with the control group were 12.9% versus 4.5% (RR, 2.85 [95% CI, 0.94–8.60]) without evidence of a difference in early neurological worsening (RR, 0.86 [95% CI, 0.58–1.25]; Table 2).

In the EVT group, procedure-related complications occurred in 6 patients (Table S2).

DISCUSSION

In this post hoc analysis of the LASTE trial, 181 patients age 80 or younger, with a prestroke score of 0 or 1 on the mRS, presenting within 6.5 hours of stroke onset with a proximal large-vessel occlusion in the anterior circulation and largest baseline infarcts (defined by ASPECTS 0–2), with no upper limit of infarct size, were included. In our population, well balanced for the baseline characteristics between allocated treatment, EVT was significantly

Table 1. Demographic and Clinical Characteristics of the Patients With ASPECTS 0–2 (Assessed by Core Lab Evaluation) at Baseline

Characteristics	Medical care (n=95)	EVT (n=86)
Age, y	72 (63–78)	72 (64–78)
Male sex	53/95 (55.8)	48/86 (55.8)
Medical history		
Hypertension	59/95 (62.1)	52/86 (60.5)
Hypercholesterolemia	25/95 (26.3)	16/86 (18.6)
Diabetes	34/95 (35.8)	29/86 (33.7)
Current smoking	17/94 (18.1)	15/86 (17.4)
Coronary artery disease	14/95 (14.7)	18/86 (20.9)
Previous stroke or TIA	14/95 (14.7)	20/86 (23.3)
Atrial fibrillation	22/95 (23.2)	19/86 (22.1)
Previous antithrombotic medications	41/95 (43.2)	49/86 (57.0)
Antiplatelet	29/95 (30.5)	34/86 (39.5)
Anticagulant	16/95 (16.8)	19/86 (22.1)
Current stroke event		
ASPECTS (core lab evaluation)		
0	19/95 (20.0)	15/86 (17.4)
1	32/95 (33.7)	26/86 (30.2)
2	44/95 (46.3)	45/86 (52.3)
Transfer to center with EVT capabilities	54/95 (56.8)	50/86 (58.1)
Prestroke mRS score		
0	79/95 (83.2)	64/86 (74.4)
1	14/95 (14.7)	21/86 (24.4)
>1	0/95 (0.0)	1/86 (1.2)
Systolic blood pressure, mm Hg	143 (125–164)	152 (131–167)
Plasma glucose level, g/L	1.4 (1.2–1.6) [n=93]	1.3 (1.1–1.6) [n=84]
NIHSS score at admission	21 (18–25)	21 (18–24)
Qualifying Imaging method		
CT	11/95 (11.6)	14/86 (16.3)
MRI	84/95 (88.4)	72/86 (83.7)
Infarction volume at baseline (core lab evaluation), mL	157 (123–202)	155 (118–213)
Occlusion site (core lab evaluation)		
ICA	51/95 (53.7)	46/86 (53.5)
M1 segment	43/95 (45.3)	38/86 (44.2)
Others	1/95 (1.1)	2/86 (2.3)
Stroke cause		
Large artery atherosclerosis	2/95 (2.1)	11/86 (12.8)
Cardioembolic	45/95 (47.4)	38/86 (44.2)
Other certain cause	6/95 (6.3)	4/86 (4.7)
Unknown	42/95 (44.2)	33/86 (38.4)
Intravenous thrombolysis	33/95 (34.7)	28/86 (32.6)
Symptom onset unknown	31/95 (32.6)	28/86 (32.6)
Symptom onset unknown with DWI-FLAIR mismatch*	20/55 (36.4)	17/47 (36.2)
Interval from last known to be well to, min		
Randomization	267 (207–345)	285 (190–357)
Qualifying imaging	167 (115–283)	181 (117–322)
Admission to center with EVT unit	252 (161–313) [n=92]	256 (154–315)
Gro In puncture	NA	309 (230–381)

Values are n/total n (%) or median (25th–75th percentiles) [no. of available patients in case of missing data]. ASPECTS indicates Alberta Stroke Program Early Computed Tomography Score; CT, computed tomography; DWI-FLAIR, diffusion-weighted imaging–fluid-attenuated inversion recovery; EVT, endovascular thrombectomy; ICA, internal carotid artery; MRI, magnetic resonance imaging; mRS, modified Rankin Scale, NA, not applicable; NIHSS, National Institutes of Health Stroke Scale; and TIA, transient ischemic attack.

*The assessment of the DWI-FLAIR mismatch profile was done by the center and was missing for 13 patients in the medical care group and 16 in the EVT group.

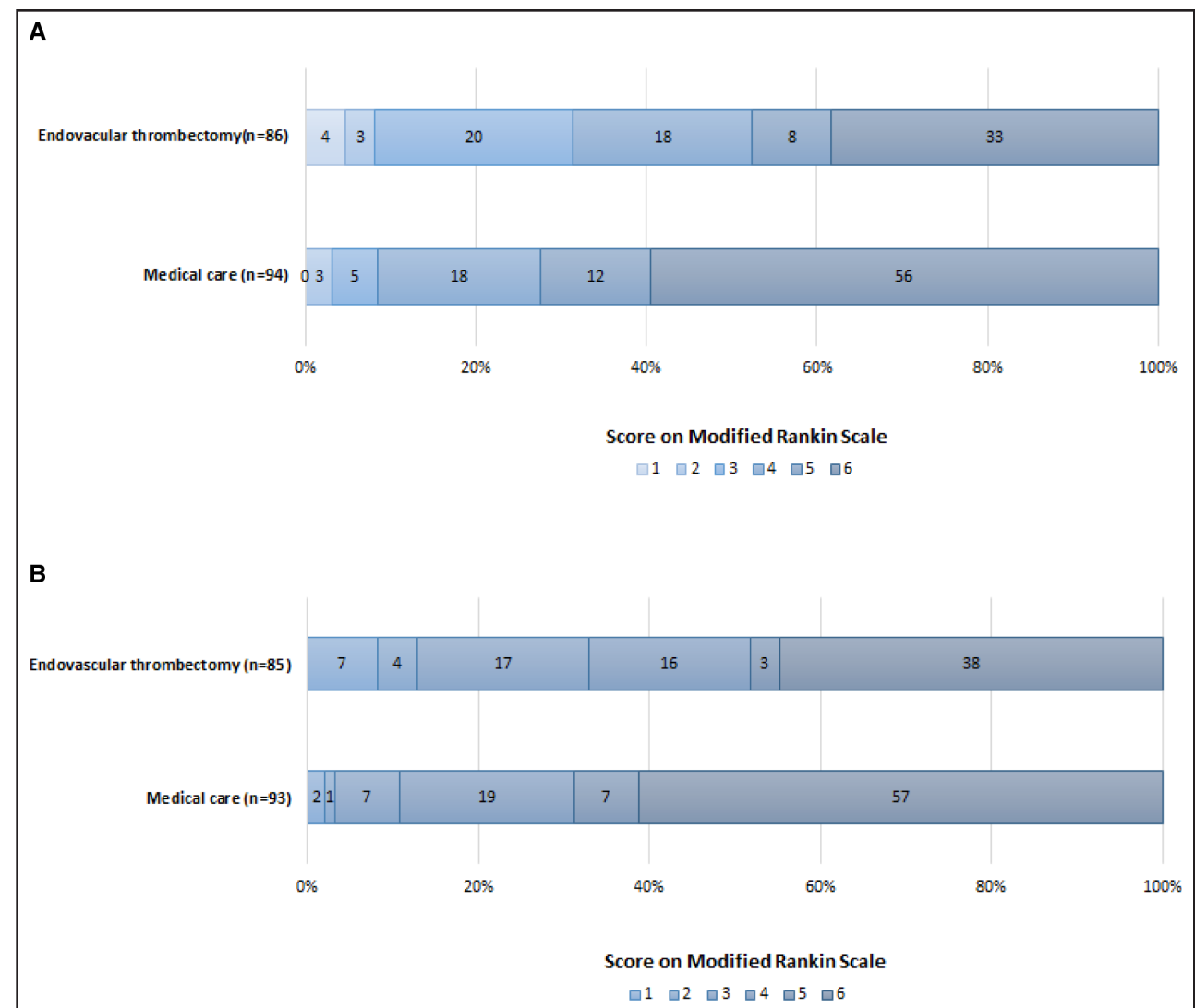


Figure 1. Distribution of the modified Rankin Scale score at day 90 and at day 180.

Distribution of the modified Rankin Scale score at day 90 (A) and at day 180 (B) according to allocated arm in patients with baseline Alberta Stroke Program Early Computed Tomography Score 0 to 2.

associated with reduced disability and mortality rates, challenging the idea of exclusion of patients solely based on their baseline infarct size.

These patients whose median core volume measures 156 cm³, substantially higher than that reported in other large-core trials conducted to date, constitute ≈50% of the entire LASTE population. This distribution of infarct sizes among LASTE patients with over 50% of ASPECTS scores in the 0 to 2 range and over 50% of patients enrolled with infarct volumes of >150 mL reflects the distribution of large baseline infarct volumes in patients with proximal large-vessel occlusion presenting in the early time window previously described in consecutive case series.¹² This in conjunction with the screening logs from LASTE, which did not reveal that patients were excluded from randomization because of too large infarct sizes, suggest that our patient population is representative in terms of large infarct values of

all patients presenting in the early time window at the participating thrombectomy centers. Although most of the patients enrolled in LASTE were selected based on MRI (>80%), the fact that LASTE enrolled patients with the largest infarcts that mirror the normal distribution of large infarcts in consecutive patients presenting in the early time window, renders the imaging modality used for patient selection (CT versus MRI) diminished in relevance. However, data from other large-core studies that used CT as the selection imaging modality will help to better clarify this issue.

It is important to note that the baseline size of infarct remains a poor prognostic factor, as showed by the rate of favorable outcome compared with smaller core.¹³ Nevertheless, EVT was significantly associated with better outcome reinforcing the concept that at least in the early time window it does not modify treatment effect.

Table 2. Efficacy and Safety Outcomes

Outcomes	Medical care (n=95)	Endovascular thrombectomy (n=86)	Treatment effect (95% CI)*	P value
Efficacy outcomes				
Modified Rankin Scale score at 90 d				
Median (25th–75th percentile)	6 (4 to 6) [n=94]	4 (3 to 6)	1.81 (1.32 to 2.47)	<0.001
0–2	3/94 (3.2)	7/86 (8.1)	2.55 (0.68 to 9.56)	0.20
0–3	8/94 (8.5)	27/86 (31.4)	3.69 (1.77 to 7.68)	<0.001
4–5	30/94 (31.9)	26/86 (30.2)	0.95 (0.61 to 1.47)	0.81
Modified Rankin Scale score at 180 d				
Median (25th–75th percentile)	6 (4 to 6) [n=93]	4 (3 to 6) [n=85]	1.69 (1.23 to 2.31)	0.001
0–2	3/93 (3.2)	11/85 (12.9)	4.01 (1.15 to 13.90)	0.016
0–3	10/93 (10.8)	28/85 (32.9)	3.06 (1.58 to 5.93)	<0.001
4–5	26/93 (28.0)	19/85 (22.4)	0.80 (0.47 to 1.34)	0.39
Decompressive craniectomy within 7 d	14/95 (14.7)	11/86 (12.8)	0.88 (0.39 to 1.93)	0.74
Early neurological improvement†	4/90 (4.4)	20/83 (24.1)	5.42 (1.93 to 15.21)	<0.001
Baseline-adjusted mean 24-h change in infarction volume (95% CI), mL	128.5 (111.5 to 145.4)	58.2 (41.0 to 75.4)	–70.3 (–94.2 to –46.35)	<0.001
Safety outcomes				
Death from any cause within 90 d	56/94 (59.6)	33/86 (38.4)	0.64 (0.47 to 0.89)	0.005
Symptomatic intracranial hemorrhage within 24 h (Heidelberg classification)	4/88 (4.5)	11/85 (12.9)	2.85 (0.94 to 8.60)	0.050
Early neurological worsening‡	38/90 (42.2)	30/83 (36.1)	0.86 (0.58 to 1.25)	0.41

Values are n/total n (%) unless otherwise as stated. NIHSS indicates National Institutes of Health Stroke Scale.

*The treatment effect was assessed using fourth different measures: (1) the generalized odds ratio and its 95% CI is provided for the ordinal shift in the score distribution on the modified Rankin Scale, indicating outcome improvement (collapsed categories 5 and 6); (2) the subdistribution hazard ratio, with its 95% CI, are provided for the incidence of decompressive craniectomy, treating mortality as competing risk; (3) the baseline-adjusted mean difference with its 95% CI are provided for the 24-hour change in infarction volume; (4) the relative risk and its 95% CI are provided for binary outcomes.

†Early neurological improvement was defined as a decrease of ≥8 points in the NIHSS score from the time of presentation to a center with endovascular thrombectomy capabilities to day 7/or discharge (whichever earlier); death within 7 days was considered as neurological worsening.

‡Early neurological worsening was defined as an increase of ≥10 points in the NIHSS score from the time of presentation to a center with endovascular thrombectomy capabilities to the day 7 or discharge (whichever earlier); death within 7 days was considered as neurological worsening.

Although the patients enrolled in LASTE achieved low overall rates of desirable outcomes regardless of treatment allocation, EVT was associated with significantly improved functional outcomes and reduced mortality, despite a numerically higher risk of sICH and procedural complications. Moreover, the observed mortality gains have not resulted in higher rates of undesirable outcomes as the difference in mortality rates between the 2 groups mainly translated into differences in rates of mRS score 0 to 3 (an outcome most patients would prefer to death¹⁴) while rates of mRS score 4 and 5 remained similar between the 2 groups (Figure S4). Unexpectedly, the magnitude of the treatment effect expressed as a number needed to treat of 3.5 (95% CI, 2.3–7.2) to achieve a lower level of disability with EVT compared with medical care alone is similar to the one seen in patients presenting with small or moderate size baseline core treated in the early time window (number needed to treat of 4 in HERMES [Highly Effective Reperfusion Evaluated in Multiple Endovascular Stroke Trials]). Our results pertaining to a subgroup of LASTE patients, mirror the overall results of LASTE and are particularly noteworthy given the limited data available on patients with ASPECTS

0 to 2 from other major large-core EVT trials in which apart from TESLA that allowed enrollment of patients with ASPECTS 2, inclusion was limited to patients with ASPECTS scores between 3 and 5. In these trials, owing to a discrepancy between investigator-adjudicated ASPECTS scores and core lab adjudicated ASPECTS scores, a small subset of patients were found based on core lab adjudication to have presented with ASPECTS scores of 0 to 2. A recent study-level meta-analysis that included these non LASTE patients with core lab adjudicated ASPECTS 0 to 2 revealed results that are in line with our findings.¹⁵

This study-level meta-analysis of 6 trials evaluating the subgroup of 360 patients with ASPECTS 0 to 2 showed that EVT improved the rates of 90-day independent ambulation (mRS score of 0–3: 26.2% versus 12.2%; RR, 2.12 [95% CI, 1.35–3.32]; *P*<0.01) and functional independence (mRS score 0–2, 9.9% [17/172] versus 3.7% [7/188]; RR, 2.78 [95% CI, 1.15–6.72]; *P*=0.02).¹⁵ Individual level meta-analyses with uniform core lab adjudication across all 6 trials should provide more clarification regarding the benefit and risks of thrombectomy in this category of patients.

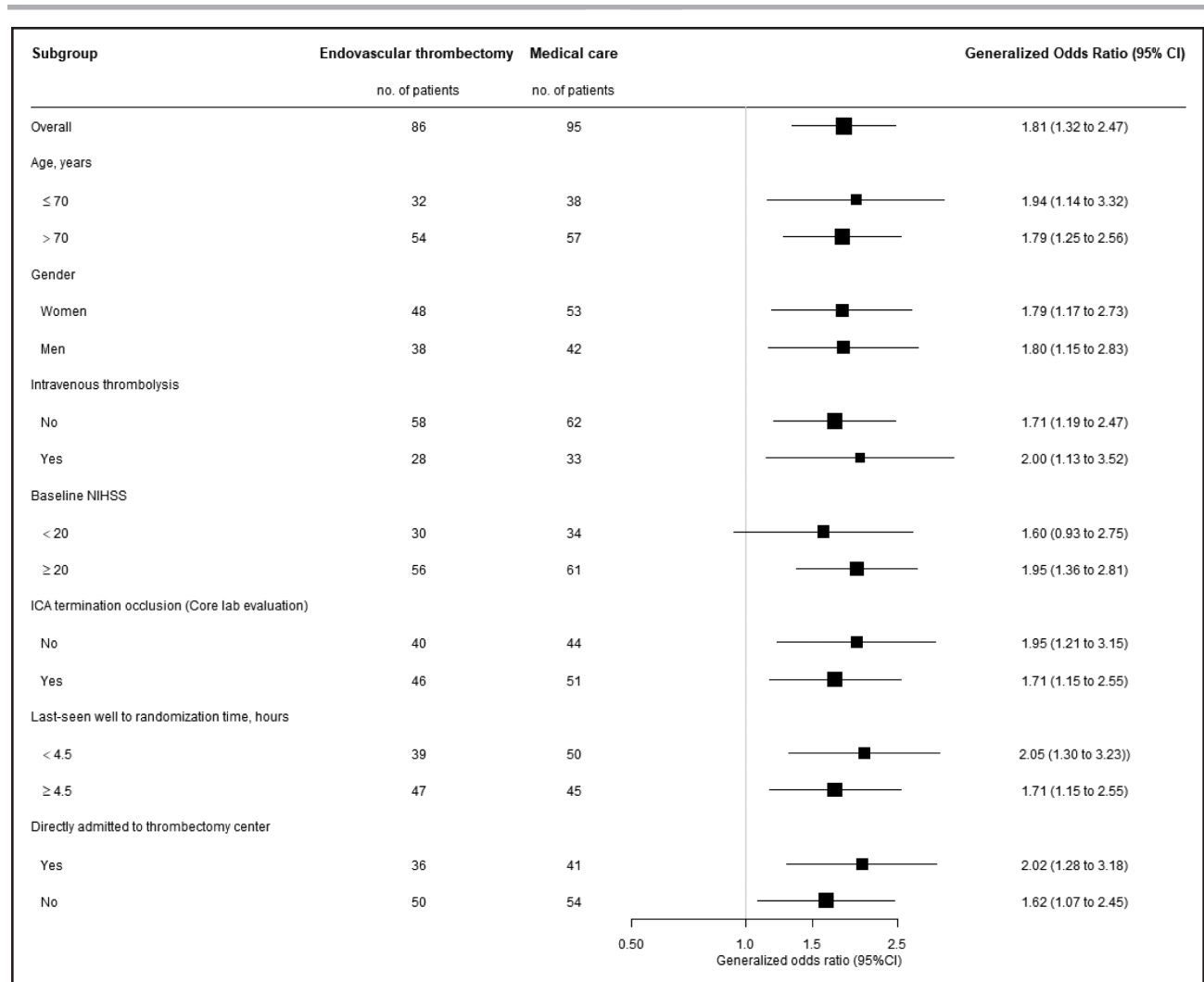


Figure 2. Treatment effect size for the primary outcome in patients with baseline Alberta Stroke Program Early Computed Tomography Score 0 to 2 according to key subgroups.

NIHSS indicates National Institutes of Health Stroke Scale.

Because the observed treatment effect (though not the overall prognosis) of EVT in patients with the largest baseline infarcts is not significantly different compared with that seen in the ASPECTS 3 to 5 of LASTE and other randomized thrombectomy trials irrespective of baseline core size, it follows that even in cases of seemingly large infarcted territory as assessed by imaging, there remains an opportunity to salvage viable brain tissue. Systematic perfusion imaging was not mandated in LASTE. Within the ASPECT 0 to 2 populations with available perfusion, the rate of patients with perfusion-diffusion mismatch (defined as mismatch ratio ≥ 1.2) was 36/73, 49%. Alternatively, there may be mechanisms underlying reperfusion that are distinct from penumbral salvage and may act in conjunction with penumbral salvage to induce the observed clinical benefit derived from thrombectomy. Such mechanisms may include recovery enhancing effects through preservation of non-neuronal tissue¹⁶ and edema reduction. Indeed, mediation analysis studies have shown that in large-core thrombectomy, edema reduction

brought about by reperfusion is a more important driver of clinical benefit than penumbral salvage.^{17,18}

In LASTE, the positive clinical impact of EVT in patients with ASPECT 0 to 2 was only studied within the context of early treatment initiation. Indeed, the median time from last known well to randomization was 267 minutes (interquartile range, 207–345) in the medical care group and 285 minutes (interquartile range, 190–357) in the EVT group. The relatively shorter treatment times in LASTE underscore the critical importance of early intervention, particularly in patients with severe strokes. This timely management likely contributed to the favorable outcomes observed, reinforcing the potential benefit of EVT in ASPECTS 0 to 2 patients who are recanalized early.

The effectiveness of EVT within a shorter time frame may have different explanations: (1) possibility that large core does not represent complete bioenergetic failure—several studies have demonstrated persistent microvascular perfusion and islands of viable neurons within diffusion-abnormal regions¹⁶; (2) the role of reperfusion

in reducing cytotoxic and vasogenic edema, which appears to be a strong mediator of benefit in large-core populations¹⁷; (3) early reperfusion may rescue penumbral tissue located at the periphery of large infarcts, and may also preserve non-neuronal structures (glia, extracellular matrix) essential for recovery.

Further study is necessary to clarify the benefit of EVT in patients with ASPECTS 0 to 2 presenting the late window.

The rate of decompressive craniectomy within the first 7 days poststroke was similar between groups (14.7% in the medical care group versus 12.8% in the EVT group; odds ratio, 0.88 [95% CI, 0.39–1.93]; $P=0.74$). There are multiple reasons behind this rather unexpected finding. Different family member attitudes towards agreeing to hemicraniectomy on behalf of their loved ones according to treatment allocation may be one of them. Nonetheless there remains a possibility that EVT did not reduce the incidence of malignant edema, possibly indicating that its beneficial effects were less due to a lower incidence of severe brain swelling and more due to its role in limiting the extent of irreversible brain injury.

Regarding sICH, the 95% CI approaches 1.0 for ICH ($P=0.050$) indicating that a larger sample may have altered the statistical significance of the sICH comparison. Indeed, an individual patient data meta-analysis (ATLAS), presented at the World Stroke Conference in 2025, pooling data from the 6 large-core studies, confirmed an association with an increased risk of symptomatic intracerebral hemorrhage: EVT: 5.5% versus MM, 2.7%; absolute risk reduction, 2.02 [95% CI, 1.13–3.59]; $P=0.017$. The increased incidence of symptomatic intracranial hemorrhage in the EVT group highlights the need for exploring future treatment strategies aiming to mitigate the development of this significant complication. Meanwhile, although the benefits of EVT in terms of functional recovery and survival are evident, clinicians must weigh these advantages against the heightened risk of hemorrhagic complications, particularly in patients with extensive infarcts.

Clinical Implications and Future Directions

Our findings suggest that in patients younger than 80 years, with a prestroke score of 0 or 1 on the mRS, presenting within 6.5 hours of last seen well, the extent of baseline infarction, as measured by ASPECTS or by volumetric analysis on CT or MRI is a prognostic factor but not a treatment effect modifier. Given the high likelihood of outcomes that some patients may perceive as undesirable even with EVT, the decision to proceed with this treatment is a complex decision that should consider patient's presumed survival preferences despite a high likelihood of significant losses in quality of life. However, a critical insight emerging from our analysis is that extent of infarct on baseline imaging, regardless of its size should not be the sole criterion for EVT eligibility.

Strengths and Limitations

This study has several strengths, including its prospective, multicenter design and exclusive focus on patients with ASPECTS 0 to 2, a population often excluded from previous trials, with comparable baseline characteristics between allocated arms. However, certain limitations must be acknowledged. The premature termination of the trial may have impacted statistical power. Furthermore, at first glance the predominance of MRI-based imaging selection could render the results of our study less applicable to patients selected on CT-based protocols. However, due to the fact that patients were not excluded from the trial based on infarct size, it is likely that our results apply to patients selected both with MRI and with CT.

Only patients aged <80 years with ASPECTS value of 0 to 2 were enrolled in the LASTE trial. As such, and given the poor prognosis for a meaningful neurological outcome conferred by both a large infarct and advanced age, the effects of EVT in octogenarians may not be generalizable to this patient population.

Conclusions

EVT improves functional outcomes and reduces mortality in patients with ASPECTS 0 to 2. These findings support the concept that in patients younger than 80 years presenting within early time window (6.5 hours) with unlimitedly large infarct (predominantly selected using MRI), the infarct size in isolation should not be used to disqualify patients from endovascular treatment.

ARTICLE INFORMATION

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Affiliations

Department of Neurology (B.L.) and Department of Neuroradiology (F.D.M.), Hôpital Fochpital Foch, Suresnes, France. Department of Neurology (C.A.) and Department of Neuroradiology (V.C.), Hôpital Gui de Chauliac, Montpellier, France. Department of Neurology and CIC 1436 (J.F.A.) and Department of Neuroradiology (C.C.), Hôpital Pierre Paul Riquet, Toulouse, France. Department of Neurology (H.H.) and Department of Neuroradiology (N.N.), Hôpital Salengro, Lille, France. Department of Neuroradiology (B.G.) and Department of Neurology (S.R.), Hôpital Central, Nancy, France. Department of Neuroradiology (G.M.) and Department of Neurology (I.S.), Hôpital Pellegrin, Bordeaux, France. Department of Neurology (M.A.) and Department of Neuroradiology (G.B.), Hôpital Bretonneupital Bretonneau, Tours, France. Department of Neurology, Hospital Universitari de Bellvitge, L'Hospitalet de Llobregat, Spain (P.C.). Department of Statistics, Economic Evaluation, Data-Management Unit, University Medical Center, Lille, France (J.L.). Cooper Neurological Institute, Cooper University Healthcare and Cooper Medical School of Rowan University, Camden, NJ (T.G.J.). Versailles Saint Quentin en Yvelines University, Suresnes, France (B.L.). INSERM U1266, Paris, France (C.A.). Imagerie Adaptative Diagnostique et Interventionnelle (IADI), INSERM U1254, University of Lorraine, Nancy, France (B.G.).

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Supplemental Material

Tables S1–S2
Figures S1–S5
CONSORT Checklist

APPENDIX

The authors' full names and academic degrees are as follows: Vincent Costalat, MD, PhD; Tudor G. Jovin, MD; J.F. Albucher, MD; Christophe Cognard, MD, PhD; Hilde Henon, MD; Nasreddine Nouri, MD; Benjamin Gory, MD, PhD; Sebastien Richard, MD; Gaultier Marnat, MD; Igor Sibon, MD, PhD; Federico Di Maria, MD; Mariam Annan, MD; Gregoire Boulouis, MD; Pere Cardona, MD; Michael Obadia, MD; Michel Plotin, MD, PhD; Romain Bourcier, MD, PhD; Benoit Guillon, MD; Sophie Godard, MD; Anne Pasco-Papon, MD; Omer F. Eker, MD, PhD; Tae-Hee Cho, MD, PhD; Guillaume Turc, MD, PhD; Olivier Naggara, MD, PhD; Stephane Velasco, MD; Matthias Lamy, MD; Frederic Clarencon, MD, PhD; Sonia Alamowitch, MD, PhD; Arturo Renu, MD, PhD; Laurent Suissa, MD, PhD; Herve Brunel, MD; Jean-Christophe Gentic, MD; Serge Timsit, MD, PhD; Chantal Lamy, MD; Cyril Chivot, MD; Francisco Macian-Montoro, MD; Charbel Mounayer, MD; Ozlem Ozkul-Wermester, MD; Chrysanthi Papagiannaki, MD; Valerie Wolff, MD, PhD; Raoul Pop, MD, PhD; Anna Ferrier, MD; Emmanuel Chabert, MD, PhD; Frederic Ricolfi, MD, PhD; Yannick Bejot, MD, PhD; Elena Lopez-Cancio, MD, PhD; Pedro Vega, MD; Laurent Spelle, MD, PhD; Christian Denier, MD, PhD; Monica Millan, MD, PhD; Juan F. Arenillas, MD; Mikael Mazighi, MD, PhD; Emmanuel Houdart, MD, PhD; Maria del Mar Freijo, MD; Alain Duhamel, PhD; Nerses Sanossian, MD; David S. Liebeskind, MD; Julien Labreuche, MSc, Bertrand Lapergue, MD, PhD; Caroline Arquizan, MD

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